

Lecture #15: Markov Chains and multistep Transitions

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Review of Markov Chains

Example

Consider a gambling game in which on any turn you win \$1 with probability $p = 0.4$ or lose \$1 with probability $1 - p = 0.6$. Suppose further that you adopt the rule that you quit playing if your fortune reaches $\$N$. Of course, if your fortune reaches $\$0$ the casino makes you stop.

Definition

Let X_n be the amount of money you have after n plays. Your fortune, X_n has the “Markov property.” In words, this means that given the current state, X_n , any other information about the past is irrelevant for predicting the next state X_{n+1} .

Definition

We say that X_n is a discrete time *Markov chain* with *transition matrix* $p(i, j)$ if

$$\mathbb{P}(X_{n+1} = j | X_n = i) = p(i, j)$$

The Gambler's Ruin

Example

Describing the Gambler's ruin in terms of a probability matrix.

$$p(0,0) = 1$$

$$p(N,N) = 1$$

$$p(i, i+1) = 0.4, \quad 0 < i < N$$

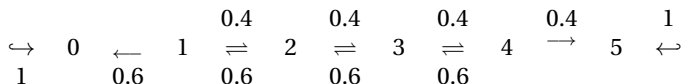
$$p(i, i-1) = 0.6, \quad 0 < i < N$$

If we make a matrix out of this we have

	0	1	2	3	4	5
0	1.0	0	0	0	0	0
1	0.6	0	0.4	0	0	0
2	0	0.6	0	0.4	0	0
3	0	0	0.6	0	0.4	0
4	0	0	0	0.6	0	0.4
5	0	0	0	0	0	1.0

Graphical Look at Gambler's Ruin

We can also view this as a collection of states and ways to move between them.



One more commonly describes a Markov chain by writing down a transition probability $p(i, j)$ with

- ① $p(i, j) \geq 0$, since they are probabilities.
- ② $\sum_j p(i, j) = 1$, since when $X_n = i$, X_{n+1} will be in some state j .

The equation is read “sum $p(i, j)$ over all possible values of j .” In words the last two conditions say: the entries of the matrix are nonnegative and each ROW of the matrix sums to 1.



More Markov Chains

Example

Let X_n be a family's social class in the n^{th} generation, which we assume is either 1 = lower, 2 = middle, or 3 = upper. In our simple version of sociology, changes of status are a Markov chain with the following transition probability

	1	2	3
1	0.7	0.2	0.1
2	0.3	0.5	0.2
3	0.2	0.4	0.4

Do the fractions of people in the three classes approach a limit?



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① Review of Markov Chains

② Multistep Transitions



What are multistep transitions?

The transition probability $p(i, j) = \mathbb{P}(X_{n+1} = j | X_n = i)$ gives the probability of going from i to j in one step. Our goal in this section is to compute the probability of going from i to j in $m > 1$ steps:

$$p^m(i, j) = \mathbb{P}(X_{n+m} = j | X_n = i)$$

To warm up, we recall the transition probability of the social mobility chain:

	Lower	Middle	Upper
Lower	0.7	0.2	0.1
Middle	0.3	0.5	0.2
Upper	0.2	0.4	0.4



Question

Your parents were middle class (state 2). What is the probability that you are in the upper class (state 3) but your children are lower class (state 1)?

Solution

Intuitively, the Markov property implies that starting from state 2 the probability of jumping to 3 and then to 1 is given by

To get this conclusion from the definitions, we note that using the definition of conditional probability,

Question

What's the probability your children are lower class given your parents were middle?

Solution

To do this we simply have to consider the three possible states for your class and use the solution of the previous problem

There is nothing special here about the states 2 and 1 here. By the same reasoning,

The right-hand side of the last equation gives the $(i, j)^{th}$ entry of the matrix p is multiplied by itself.

To explain this, we note that to compute $p^2(2, 1)$ we multiplied the entries of the second row by those in the first column:



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$$\begin{pmatrix} \cdot & \cdot & \cdot \\ 0.3 & 0.5 & 0.2 \\ \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} 0.7 & \cdot & \cdot \\ 0.3 & \cdot & \cdot \\ 0.2 & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot \\ 0.40 & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}$$

If we wanted $p^2(1, 3)$ we would multiply the first row by the third column



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$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} \cdot & \cdot & 0.1 \\ \cdot & \cdot & 0.2 \\ \cdot & \cdot & 0.4 \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & 0.15 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}$$

When all of the computations are done we have

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix} \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix} = \begin{pmatrix} 0.57 & 0.28 & 0.15 \\ 0.40 & 0.39 & 0.21 \\ 0.34 & 0.40 & 0.26 \end{pmatrix}$$



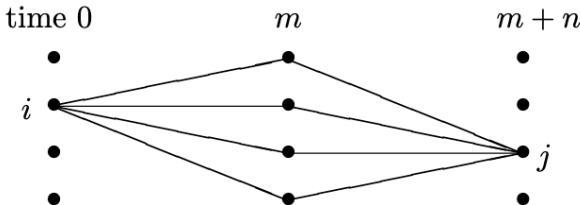
Theorem

The m step transition probability $\mathbb{P}(X_{n+m} = j | X_n = i)$ is the m^{th} power of the transition matrix p .

The key ingredient is proving the **Chapman-Kolmogorov equation**

$$p^{m+n}(i, j) = \sum_k p^m(i, k) p^n(k, j).$$

Why is this true? To go from i to j in $m + n$ steps, we have to go from i to some state k in m steps and then from k to j in n steps. The Markov property implies that the two parts of our journey are independent



Your Turn

Example (Gambler's Ruin)

Suppose for simplicity that $N = 4$ in Example 1.1, so that the transition probability is

	0	1	2	3	4
0	1.0	0	0	0	0
1	0.6	0	0.4	0	0
2	0	0.6	0	0.4	0
3	0	0	0.6	0	0.4
4	0	0	0	0	1.0

Compute p^2 .

